## **S**peak **L**ike **A** **M**athematician:

reflection:

line of reflection:

glide reflection:

line symmetry:

line of symmetry:

## Coordinate Rules for Reflections

**In the x-axis: (a, b) → (____________, ____________)** **In the y-axis: (a, b) → (____________, ____________)**

**In the line y = x: (a, b) → (____________, ____________)** **In the line y = −x: (a, b) → (____________, ____________)**

**Reflection Postulate: a reflection is a ____________________ ____________________.**

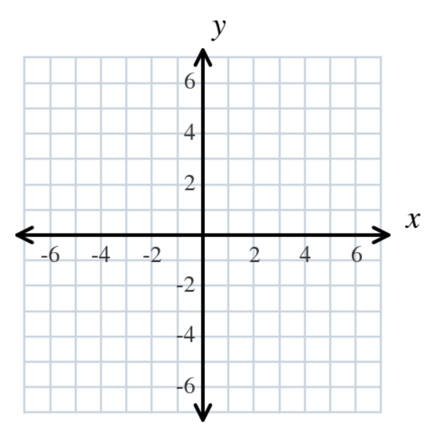
## Example 1 – Reflect in a Vertical Line

Graph △DEF with D(2, 4), E(6, 3), and F(3, 1) and its image after a reflection in the line x = 4.

a) How far is D from the line x = 4? Where does D′ land?

b) Find the coordinates of the image.

c) Write a rule for a reflection in the line x = 4.

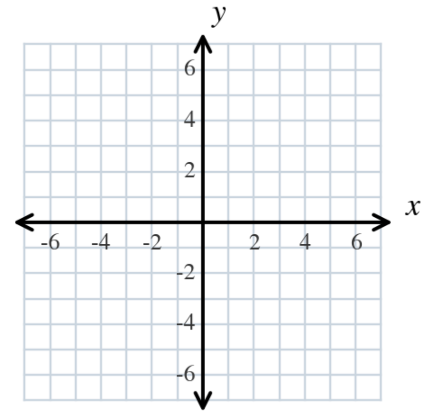


## Example 2 – Reflect in the Line y = x

Graph $\overline{MN}$ with M(−2, 3) and N(1, 4) and its image after a reflection in the line y = x.

a) Find the coordinates of the image.

b) Is MN = M′N′? Explain.




## Example 3 – Reflect a Point in Each Line

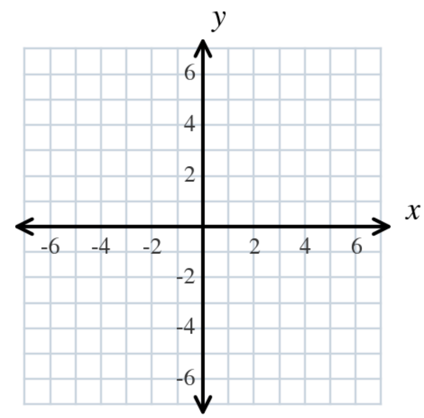
Find the image of P(5, −2) after a reflection in each line.

a) the x-axis

b) the y-axis

c) the line y = x

d) the line y = −x




## Example 4 – Glide Reflection

Graph △ABC with A(−6, 3), B(−3, 4), and C(−2, 1) and its image after the glide reflection.

**Translation:** (x, y) → (x + 8, y)

**Reflection:** in the x-axis

a) Find the coordinates after the translation.

b) Find the coordinates after the reflection.

c) Why is the order of the two steps not important here?

## Line Symmetry

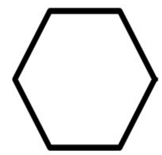
**A figure has line symmetry when the figure can be mapped ____________ ____________ by a reflection in a line.**

**That line is a line of ____________________. A figure can have ____________________ than one.**

## Example 5 – Count the Lines of Symmetry

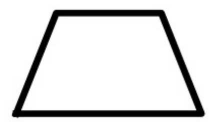
How many lines of symmetry does each figure have?

a) regular hexagon

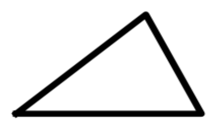


b) rectangle (not a square)

c) isosceles trapezoid



d) scalene triangle



## Example 6 – Reflect in a Horizontal Line

Graph $\overline{QR}$ with $Q(-4, 5)$ and $R(-1, 2)$ and its image after a reflection in the line $y = 1$.

a) How far is Q above the line $y = 1$? Where does $Q'$ land?

b) Find the coordinates of the image.

c) Write a rule for a reflection in the line $y = 1$.